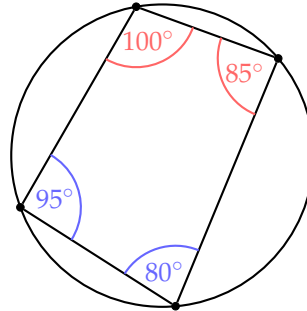


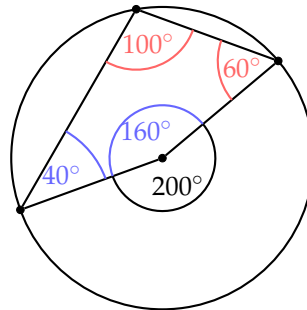
Answers

Q1.) *Note: These solutions each show **one** valid method to prove the answers for the problems. For each problem, there are alternative methods that can be used.*

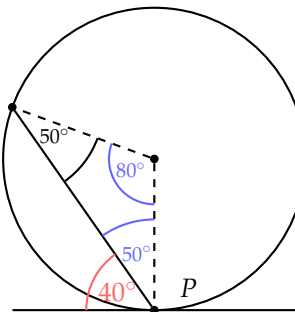
- (a) Opposite angles in a cyclic quadrilateral add to 180° . Therefore, we get $x = 95^\circ$ and $y = 80^\circ$:



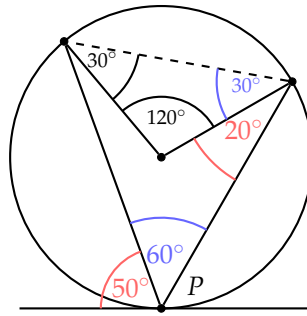
- (b) The angle at the centre is double the angle at the circumference, giving an angle of 200° . Then, since angles around a point add to 360° , $x = 160^\circ$. Since angles in a quadrilateral add to 360° , $y = 40^\circ$:



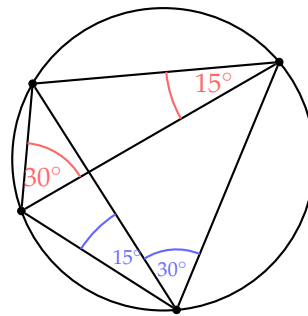
- (c) The radius and tangent meet at 90° , meaning that x and 40° add to 90° . Therefore, $x = 50^\circ$. Since the triangle is isosceles (formed by two radii) we find the opposite angle to also be 50° . Since angles in a triangle add to 180° , it follows that $y = 80^\circ$:



- (d) The alternate segment theorem tells us that $x + 20^\circ = 50^\circ$. Therefore $x = 30^\circ$. Since the smaller triangle is isosceles (formed by two radii) the angle opposite x must also be 30° . Since angles in a triangle add to 180° , the remaining angle is 120° . The angle at the centre is double the angle at the circumference, and therefore $y = 60^\circ$:



- (e) Since angles subtended at the circumference in the same segment are equal, we immediately find that $x = 30^\circ$ and $y = 15^\circ$:

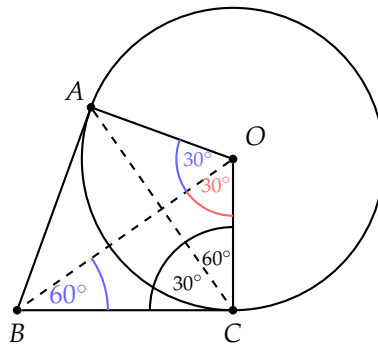


- (f) To start with let's focus on x . We can see that the two triangles OAB and OCB are **congruent** and therefore $x = 30^\circ$. To verify that they are congruent, it's easiest to either use the SSS rule or facts about kites (see Q3). Either way you'll need the theorem that tangents from the circumference to the same point have the same length.

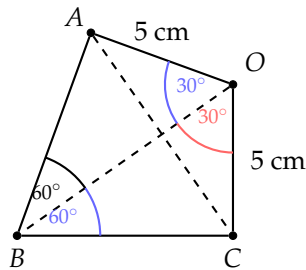
As for y , the line segment OB is a perpendicular bisector of the line segment AC . We can justify this in a few different ways: the easiest way is to remember that the diagonals of a kite meet at right-angles. We'll then need to use the following facts:

- Angles in a triangle add to 180° .
- Radii and tangents meet on the circumference at right-angles.

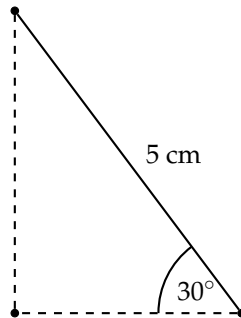
Applying these, we can work out a sequence of angles leading to the value of y being 60° as shown below:



- Q2.) We have two opposite angles of 100° and 160° , which add to 260° . A circle theorem tells us that opposite angles **must** add to 180° in a cyclic quadrilateral, so this quadrilateral is not cyclic. *Note: You can use the other pair of opposite angles instead to get the same answer.*
- Q3.) The quadrilateral is a kite due to having two pairs of opposite sides with equal lengths.
- Q4.) To make this easier to view, we can isolate the quadrilateral, noticing that it is broken into four different right-angled triangles:



To calculate the areas, there are some missing lengths we need to find. For example, two of the triangles are as follows:



We'll need to find **both** the base (adjacent) and the height (opposite) to get the area of both of these. First,

$$\sin(30^\circ) = \frac{\text{opposite}}{5}$$

$$\therefore \text{opposite} = 5 \times \sin(30^\circ) = 2.5$$

Then,

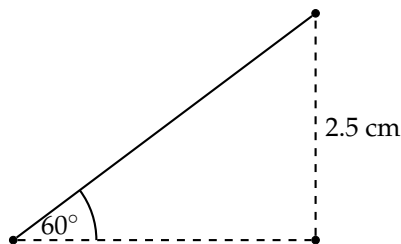
$$\cos(30^\circ) = \frac{\text{adjacent}}{5}$$

$$\therefore \text{adjacent} = 5 \times \cos(30^\circ) = \frac{5\sqrt{3}}{2} \approx 4.330127$$

Therefore,

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times \frac{5\sqrt{3}}{2} \times 2.5 \\ &= \frac{25\sqrt{3}}{8} \approx 5.41266 \end{aligned}$$

Now, we use this to deal with the remaining triangles. Using the opposite side we've just calculated, we get:



To get the area here we only need the base (adjacent), which we can use tan to calculate:

$$\tan(60^\circ) = \frac{2.5}{\text{adjacent}}$$

$$\therefore \text{adjacent} = 2.5 \times \tan(60^\circ)$$

$$= \frac{5\sqrt{3}}{2} \approx 4.33013$$

Therefore, the area of one of the larger triangles is given by

$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times \frac{5\sqrt{3}}{2} \times 2.5 \\ &= \frac{25\sqrt{3}}{8} \approx 5.41266\end{aligned}$$

Note that the diagram is not to scale here, hence the larger-looking triangles having the same area as the smaller-looking triangles. With all of the triangles have the same area of

$$\frac{25\sqrt{3}}{8}$$

the total area of the quadrilateral is therefore given by four times this, that is,

$$\text{Total Area} = \frac{25\sqrt{3}}{2} \approx 21.7 \text{ (3 s.f.)}$$